Grant Keefe

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Canadian Citizen

Education

Queen's University, BAS in Mechatronics and Robotics Engineering — Sept 2021 - May 2026

- Principal's Scholar, Dean's Scholar: GPA 3.6/4.3 (2nd year GPA 3.5 – 3rd year GPA 3.95)
- **Coursework:** Microprocessor Systems, Computer Networks, Mobile Robotics, Integrated Circuits

Professional Experience

R&D Contractor, Dominion Dynamics – Kingston, ON — Sept 2025 – May 2026

- Working on building persistent Arctic sensing nodes designed to serve both military and civilian purposes.

Research Assistant, Ingenuity Labs Research Institute – Kingston, ON — Sept 2025 – May 2026

- Working on submitting an RA-L IEEE paper for my 4th year thesis, on the use of novel sensor modalities for anomaly detection in Maritime environments.

SAR Robotics Researcher, Ingenuity Labs Research Institute – Kingston, ON — Sept 2024 – Sept 2025

- Wrote the full grant application and secured $700,000 in funding and institutional support to investigate fixed-wing UAV applications for maritime search and rescue.
- Designed the mechanical and electrical systems for a waterproof fixed-wing UAV, conducted flight testing and controller tuning, validated long-range communications links, and developed the anomaly-detection sensor processing pipeline.

Electrical Engineering Co-op, GeoSpectrum Technologies Inc. – Halifax, NS — May 2024 – Sept 2024

- Test engineer in R&D department designing and building methods for testing and troubleshooting complex systems. In depth design work done in SolidWorks and AutoCAD Electrical.

Teacher's Assistant, Queen's University – Kingston, ON — Dec 2023 – May 2024

- Responsible for trouble-shooting any hardware/software problems encountered by students throughout the course. Helped students successfully integrate ROS2 into their robot designs and take their first steps towards full autonomy.

Junior Automation Engineer, Enginuity Inc. – Halifax, NS — May 2023 – Sept 2023

- Designed and tested control panel for water mixing and dosing system. Design work done in SolidWorks and AutoCAD Electrical.

Projects

Thrust Vectored Rocket Design — Project Page

- Designed a full preliminary rocket prototype featuring thrust-vectoring gimbal, four-servo canard box with encoder feedback, a deployable nose-drogue for passive descent stabilization, and a spring-loaded parachute hatch, focusing on robust control in underactuated flight regimes.
- Developed integrated aerodynamic and mechatronic subsystems to enable active stability during powered ascent and controlled tail-down orientation during descent, forming the foundation for future work toward powered vertical landing.

Ball Gimbal — Project Page

- Designed and built a 2-DOF ball gimbal for a 400 g UAV sensor payload, completing all mechanical work (carbon fiber + 3D-printed parts), calculating inertia from STL models, and validating dynamics through Simulink before firmware integration.
- Led the full interdisciplinary development process, including three iterations of a custom gimbal control board, strengthening skills in CAD, embedded systems, manufacturing, and controls.

Unmanned Aerial System Competition — Project Page

- Led major technical development across avionics, power systems, and integration: designed and assembled

6S5P lithium-ion battery packs, created the team's first power distribution board, built the iron-bird test system, conducted full wiring and subsystem integration, and contributed to structural assembly, electrical validation, and flight-test planning.

- The aircraft earned 1st place in Prototype Realism and 2nd place in Pitch Presentation at the 2024 AEAC competition.

Micro Air Vehicle Competition Project Page

- Designed and built the complete competition drone including all avionics, sensors, airframe integration, and a battery hot-swap system; collaborated with the team to develop and validate the full mission pipeline across Gazebo simulation and real-world testing.
- The team placed 1st among undergraduate teams at the competition.

Extracurricular

President, Queen's Aerospace Design Team May 2023 – Sept 2024

- Responsible for managing a team of 60 students across seven sub-teams with a total operating budget of $40,000.
- Participated in weekly meetings for individual sub-teams, developing skills as a systems engineer

Software/Electrical Lead, Queen's Aerospace Design Team Sept 2022 – May 2023

- As a Software Lead utilized C and data structures to develop a route optimization algorithm and implemented a raspberry pi as an offboard controller sending serialized commands to fight controller.

Lacrosse Team, Queen's University Sept 2022 – Sept 2025

- 15-20 hour a week commitment

Technical Skills

Programming: Git, MATLAB, Simulink, ROS2, Gazebo, C, C++, Python, Arduino

Mechanical Design: SolidWorks + Simulation, FDM 3D Printing, SLA Printing, Laser Cutting

Electrical Design: Ki-CAD, AutoCAD Electrical, STM32 cube mx, Surface mount device (SMD) soldering, Signal probing and analysis